

A short shallow-water project

From a physical question to a controlled numerical experiment

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The aim: one small experiment, one clear result

Work in groups of two to investigate a question of your choice with the shallow-water model.

Keep the scope deliberately small

Run a baseline and at least one controlled variation. Change one main factor and support your conclusion with a quantitative diagnostic.

- Start with an assumption based on a physical process.
- Use two or three well-chosen cases; more simulations are not the goal.
- An animation can reveal a pattern, but it is not sufficient evidence.

Model scope

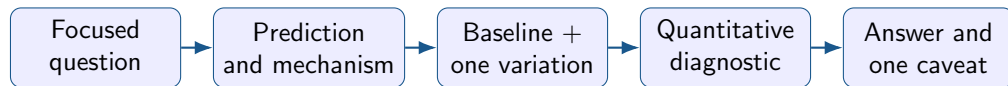
The domain remains fully wet: this is not a model of run-up, inundation, breaking, or coastal damage.

Choose a control that matches your question

Category	Examples of controls
Initial state	amplitude, radius, position, pulse shape
Bottom depth	uniform depth, shelf depth or width, ridge, input map
Wind forcing	stress, direction, duration, spatial pattern
Domain and grid	L_x , L_y , aspect ratio, N_x , N_y
Rotation	constant Coriolis parameter f
Dissipation	Rayleigh damping timescale

Domain size and resolution are different controls: changing L_x at fixed N_x also changes Δx .

Turn the question into a controlled comparison



Example

How does wind duration affect eastern-wall setup? Compare identical 3-hour and 6-hour events, then measure the maximum mean surface elevation in the four easternmost cells.

- Keep all relevant settings fixed except the chosen control.
- Record grid spacing, timestep, duration, units, and case parameters.

Workflow, submission, and assessment

Use the available time

- **Toolbox demonstration:** inspect setup recipes and choose a feasible direction.
- **Project block 1:** agree on the question and diagnostic; run the baseline and first variation.
- **Project block 2:** analyse, interpret, peer-review, and run the whole report notebook again.

Submit one executed notebook

Start from `project_report_template.ipynb`, rename it with your group identifier, and keep the evidence visible.

Assessment: G or U

A passing report has a focused question and hypothesis, a controlled comparison, quantitative evidence, a physical interpretation, one relevant caveat, reproducible settings, and a contribution statement. There are no points or weighted criteria.